

# PS 26132 & PS 26102

## The Honest Build Assessment

What's genuinely strong, what's genuinely weak,  
what could sink the demo, and how much is realistically buildable

26132 — Market linkages & price discovery for farmers (Govt. of Maharashtra / MSInS)

26102 — Anomaly, fraud & inefficiency detection in MPLADS (MoSPI)

Prepared for GlobeX — 2 September 2026

### 26132

fit rank 1 / 229   coverage 9/9

4 gaps to close

### 26102

fit rank 31 / 229   coverage 9/9

3 gaps to close

## The one-line verdict on each

### 26132 — build it, but close the gaps first

The best-measured fit GlobeX has against any of the 229 statements, and the escrow layer answers the exact fear (“payment reliability”) the problem statement names in its own text. Three specific things are not demo-ready yet: a stub data path inside counterparty matching, a forecaster whose headline number contradicts its own benchmark file, and an escrow flag that is off by default.

### 26102 — the strongest match to our best-validated asset, with one hard rule

Ranks 31st on raw similarity but is the closest match in the whole dataset to GlobeX’s single strongest, most rigorously validated component: the anomaly/risk stack. The entire pitch stands or falls on one discipline: never present the 0.98 F1 trained on trade data as a result on MPLADS data. Retrain on the scheme’s own public data and report that number, whatever it is.

## 1 What's genuinely good

### PS 26132 — Market linkages & price discovery

Across a four-profile TF-IDF similarity sweep run against all 229 statements, 26132 ranks first on the profile that matters most — product shape — by a wide margin: 65% clear of whatever sits second. Every one of the nine things the problem statement asks for (price visibility, sale-window advice, verified matching, quality/grading, lot creation, logistics, payment tracking, dispute handling, transparent records) maps to something GlobeX already has a working implementation of.

Most competing teams will build a price dashboard with a buyer directory. The problem statement itself names “payment reliability” as the thing farmers actually fear, and a dashboard does not fix that — protecting the transaction does. The escrow contract already implements fund-on-acceptance, condition-gated release, a dispute state that freezes funds instead of losing them, and a write-intent-before-chain pattern that never marks a transfer complete without a confirmed on-chain receipt — the kind of engineering discipline a technical judge notices.

The MCDA destination-ranking engine (explainable, 8 weighted dimensions, backtested at  $\rho = 0.884$ , precision@5 of 0.8 on a real 48,445-row dataset) becomes the sale-window recommender almost without modification. And the one dataset this idea absolutely needs — mandi price and arrival data — is genuinely public and reachable today through a documented API ([data.gov.in](https://data.gov.in/) / AIKosh), not locked behind a request form.

### PS 26102 — Anomaly detection in MPLADS

This is the closest match in the entire dataset to GlobeX's strongest asset. The trade anomaly classifier is trained, chronologically validated, and performing at 0.98 F1; the unsupervised risk ensemble (Isolation Forest + GRU autoencoder) exists precisely because trade data has no labelled fraud — exactly the condition public-fund irregularity data is in. Sanctioned-work expenditure records are structurally the same object as trade transactions (an amount, a date, a category, a counterparty), so the rolling-statistics and peer-deviation features transfer with minimal rework, and TreeSHAP already attaches an explanation to every flag — non-negotiable when the subject is public money.

The USP here is unusually strong and hard to copy: anchoring the flag itself (model version, input snapshot hash, score, officer disposition) to the evidence ledger, so the oversight trail is as tamper-evident as the spending it oversees. Paired with a deliberate refusal to equate “anomalous” with “fraudulent” — the existing risk documentation already makes this distinction — the pitch is restrained in exactly the way a domain involving public accusation needs it to be.

MPLADS data is also genuinely public: the scheme runs its own dashboard ([mplads.gov.in](https://mplads.gov.in/), and a newer MoSPI dashboard at [mplads.mospi.gov.in](https://mplads.mospi.gov.in/) — run by the same division that owns this PS), [data.gov.in](https://data.gov.in/) carries MPLADS-tagged datasets, and third-party aggregators like Dataful publish cleaned state/constituency-wise fund-release and expenditure figures. Getting real numbers to train and demo against is not the blocker here.

## 2 What's genuinely weak

### PS 26132

| Weakness                                                 | What's actually true                                                                                                                                                                                                                                                                                               | Why it matters                                                                                                                                        |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Counterparty matching has a stub path                    | <code>counterparty_controller.py</code> still contains an <code>hashlib.md5</code> -seeded fake-data generator on at least part of its matching path. A separate, genuine endpoint (company directory: TF-IDF + cosine + valuation blending) is real but not what “verified buyers” would naturally route through. | A judge who asks for a live match on real-looking data can hit the fake path by accident.                                                             |
| No public dataset for FPOs or buyer credentials          | No equivalent of the Agmarknet price feed exists for farmer-producer-organisation directories or buyer verification records.                                                                                                                                                                                       | The matching engine's “ground truth” has to be synthesised, and unlabelled synthetic data looks exactly like a stub-data accusation from the outside. |
| Forecaster headline number contradicts its own benchmark | The README claims 8.42% MAPE; the repo's own <code>benchmark_comparison.csv</code> shows the deployed Dual-Head GRU actually scoring 54.49% MAPE on demand — worse than a plain moving average (29.42%).                                                                                                           | The single most dangerous line in the project if a judge cross-reads the README against the repo.                                                     |
| Escrow is off by default                                 | <code>ESCROW_ENABLED</code> defaults to <code>false</code> , and only one migration file exists in the whole repository — the schema is not version-controlled.                                                                                                                                                    | The best differentiator in the project is a flag flip and an untested schema away from actually running live.                                         |

### PS 26102

| Weakness                                       | What's actually true                                                                                                                            | Why it matters                                                                                                                                                                                |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The headline accuracy is from the wrong domain | 0.98 F1 was measured on trade-anomaly data, never on MPLADS expenditure data. There is currently no retrained result on the actual scheme data. | This is the single biggest risk on this PS. Presenting 0.98 as if it were an MPLADS result is a conflation a judge will catch immediately, and it will discount everything else in the pitch. |

| Weakness                                    | What's actually true                                                                                                                                                                                                                                                               | Why it matters                                                                                                                                                                     |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Work-level detail is not cleanly structured | The public dashboards give fund-release and expenditure totals well; individual sanctioned-work records with cost estimates and execution status are less uniformly structured and partly locked in per-constituency HTML tables rather than a clean bulk API.                     | Several of the PS's specific asks (cost-estimate deviation, duplicate/overlapping works) need work-level granularity, which takes real scraping/cleaning effort to get to.         |
| No ground truth for fraud, by design        | There is no labelled "this was fraud" dataset for public works, just as there is none for trade — which is why the unsupervised ensemble exists, but it also means there is no way to report a precision/recall number against real fraud, only against synthetic or proxy labels. | Anomaly $\neq$ fraud has to be said out loud in the pitch, not discovered by the judges; otherwise it reads as an accuracy gap rather than a deliberate, defensible design choice. |

#### The two things to fix before anyone else sees either pitch

For 26132: correct the README's forecasting claim now — either retrain and report honestly, or switch the headline number to the XGBoost quantile forecaster's real performance and drop the GRU MAPE figure entirely. For 26102: retrain the anomaly/risk stack on real MPLADS data (even a modest slice) before finals, and report that number as the headline, not the 0.98 trade figure. Both are a handful of focused hours, not new capability.

### 3 Challenges, and exactly how to tackle each one

#### PS 26132

| Challenge                                                                                       | Risk if left alone                                                               | How to tackle it                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Government price data is messy — inconsistent commodity names, missing days, non-standard units | Dirty ingestion makes the forecaster look wrong live                             | Build a normalisation layer early: canonical commodity/unit mapping, a gap-fill policy, a validation pass that flags suspicious jumps before they reach the forecaster.                             |
| Stub logic inside counterparty matching                                                         | Live demo can hit fake data by accident                                          | Route “verified buyer” explicitly through the real company-directory endpoint; if the stub path stays for anything, label it unmistakably as a placeholder.                                         |
| No real FPO/buyer directory                                                                     | Matching demo runs on synthetic data that looks bad if presented as real         | Build a clearly-labelled pilot seed set (50–150 records styled on real APMC/cooperation structures) and say so in the pitch: “pilot seed data; onboarding real FPOs is the first 90-day milestone.” |
| Forecaster overclaiming / domain shift                                                          | GRU was validated on trade-corridor data, not mandi series                       | Lead with the XGBoost quantile forecaster (F2), recalibrated on actual ingested mandi series; present every recommendation as a range with a confidence band, never a bare number.                  |
| Escrow reliability under demo conditions                                                        | A live chain call failing in front of judges is worse than never having built it | Flip <code>ESCROW_ENABLED</code> on, dry-run the happy path and dispute path repeatedly on testnet, add schema migrations now, keep a recorded fallback run.                                        |

#### PS 26102

| Challenge                              | Risk if left alone                                                                                                | How to tackle it                                                                                                                                                                                                                                                                                                  |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No MPLADS-trained result yet           | The only number in hand (0.98 F1) is from the wrong domain                                                        | Pull a real slice of MPLADS expenditure/works data (mplads.gov.in, MoSPI dashboard, data.gov.in, Dataful), engineer the same rolling-stats/peer-deviation features, retrain, and report whatever number results — even if it is far lower than 0.98. A modest honest number here beats a borrowed impressive one. |
| Work-level records are semi-structured | Cost-estimate deviation and duplicate-work detection need per-work granularity that isn’t cleanly exposed in bulk | Scope the demo to a small number of constituencies where per-work detail can realistically be scraped and cleaned in the time available, rather than attempting national coverage.                                                                                                                                |

| Challenge                                         | Risk if left alone                                                                                                       | How to tackle it                                                                                                                                                                                |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anomaly vs. fraud framing                         | An unprepared team either overclaims (“we detect fraud”) or looks weak when pressed on accuracy against real fraud       | State the distinction proactively in the pitch: the system flags statistical irregularity for human review; it does not accuse. This is a documented design position, not an improvised excuse. |
| Sensitivity of accusing public officials/agencies | A false or careless flag against a real MP’s fund or agency is reputationally serious, unlike a commercial trade anomaly | Keep every flag routed to a human-in-the-loop disposition step, never auto-publish a flag, and make the evidence-ledger anchoring cover the review decision too, not just the raw score.        |

## 4 How much can realistically be built — a scope plan

### PS 26132

**Must-have:** ingestion of Agmarknet/AIKosh mandi data for 2–3 states and a handful of commodities, cleaned and normalised; sale-window recommendation via the XGBoost quantile forecaster presented as a price band; buyer matching through the real (non-stub) endpoint against the pilot seed dataset, with a visible trust score; escrow happy path live on testnet with the evidence-ledger entry visible; a clickable front-end flow start to finish.

**Should-have:** dispute path demoed live, not just described; document/grading verification against a sample certificate and lot record; the audit trail exposed in the UI; coverage widened to 4–5 states once the core flow is solid.

**Could-have (genuine stretch, do not commit):** an image-based grade classifier if a labelled photo dataset can be assembled in time; a vernacular or WhatsApp-bot front end; multi-buyer competitive offers.

**Won't-have — say so upfront:** live eNAM transaction data or GSTN/bank payment-rail integration; nationwide coverage at launch; a trained (rather than rules-based) grading model unless the stretch goal is actually reached.

### PS 26102

**Must-have:** a real, cleaned MPLADS dataset for a handful of constituencies (fund release, expenditure, work-level records where scrapeable); the anomaly classifier and unsupervised risk ensemble retrained on that data with an honestly reported score; TreeSHAP-style explanation attached to every flag; a monitoring dashboard showing flagged works with their contributing factors; the flag-anchoring-to-ledger mechanism working end to end for at least one flagged case.

**Should-have:** duplicate/overlapping-work detection via TF-IDF similarity across constituencies; cost-estimate peer-deviation scoring; a human disposition workflow (reviewer marks a flag reviewed/dismissed, and that decision is itself anchored).

**Could-have (genuine stretch):** coverage extended across more states/constituencies; a trend/early-warning view that aggregates flags by implementing agency over time.

**Won't-have — say so upfront:** any claim of detecting or proving fraud (only irregularity worth review); real-time ingestion tied to live government systems rather than periodically refreshed public data; national-scale coverage at launch.

#### On “how much can we make” as a competitive question, for both

Conditional on closing the gaps listed above, both are top-tier entries on paper: 26132 for the measured #1 fit and a differentiator the problem statement names outright; 26102 for pairing GlobeX's single best-validated component with an accountability mechanism (ledger-anchored flags) that is genuinely hard to copy in a hackathon window. Left unfixed, in both cases the failure mode is identical — a sharp judge finds the domain-mismatched number or the stub path in the first ten minutes of questions, and a top-tier submission reads as an average one. The idea and the fit do not change; the trust does. In both cases the gap between those two outcomes is single-digit hours of focused, unglamorous work, not new capability.

### Bottom line

26132 is the stronger overall submission — best fit in the dataset, and a differentiator the PS names in its own text. 26102 is the stronger showcase for GlobeX's actual ML rigor and pairs unusually well with the blockchain layer through ledger-anchored flags, but it carries one hard, non-negotiable discipline: never show the trade-trained 0.98 F1 as an MPLADS result. Whichever is filed, walk in having already



fixed the domain-mismatched numbers, rerouted matching or retrained on real data as applicable, and turned on and tested the chain layer — before touching a single new feature.